EEG processing for NF systems

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Instrucciones:

1. Use one EEG recording that you and your team recorded in the lab, and that is related to your challenge. The recording must be of at least 3 minutes, and it must include two different conditions (e.g. rest, attention, meditation, etc...). Alternatively, you can use an EEG signal from a database, but with the same duration, and conditions criteria. The experimental conditions must be related to your challenge.

2. Load your signal into EEGLAB, and show the sample rate, number of channels, and duration of your recording.

3. Use a band-pass filter to remove drift, and powerline noise. Specify the frequency ranges used and explain your choice.

4. Estimate the neuromarker (across time) you are interested in monitoring for your NF system, for all the duration of your EEG recording. Explain why and how you estimated such neuromarker. Also, if your neuromarker is the result of the combination of multiple channels, explain this process as well.

5. Generate a report to document the previous steps, including a conclusion related to this analysis in the context of your challenge. Also make sure to include the following figures:

(1) Raw EEG signals (across time), identifying the times of each condition.

(2) EEG signals (across time) after the band-pass filter, identifying the times of each condition.

(3) Neuromarker (across time), identifying the times of each condition.

Base de datos: https://ieee-dataport.org/open-access/dasps-database

Las condiciones que presenta son los siguientes

Organización de los datos

Nivel	Detalle
Sujetos	23 participantes (S01-S23)
Situaciones	6 escenarios ansiógenos por sujeto
Canales EEG	14 electrodos
Frecuencia	128 Hz
Segmentación	Ventanas de 1 segundo = 128 muestras

```
fs = 128;
%Datos del paciente uno
path = 'Raw data mat/S01.mat';
paciente_1_mat = load(path); %struct. Tambien lo tenemos con las grabaciones seguidas
labels = paciente_1.labels
```

```
labels = 12×2
     1     8
     1     8
     3     8
     3     8
     1     8
     1     8
     3     5
     3     5
     3     6
     3     6
     1     7
     1     7
     ⋮
```

```
valence = labels(:, 1);
arousal = labels(:, 2);

% grabaciones impar significa cuando recitaba el psicologo la situacion de
% estres, grabaciones par cuando la persona recitaba lo que se le conto
datos_p1 = paciente_1.data;
size(datos_p1)
```

```
ans = 1×3
        14        1920          12
```

El tamaño es de 14 eelctrodos x 1920 x 12

- 14 derivaciones
- 1920 muestras = 15 seg
- Las 6 situaciones con fases

1. Escucha: par
2. Recall: impar

Para usar eeglab correctamente contrajimos las 12 situaciones en una lectura seguida

Pasamos de 12 grabaciones de 1920 muestras => 15 segundos

Una sola grabacion de 23040 muestras => 3 minutos

```
EEG = pop_loadset('C:\Users\jorge\Documents\Tec\Uni\6. Sexto Semestre\EEG_Luzma\EEG
processing for NF systems\S01_eeglab.set')
```

```
pop_loadset(): loading file C:\Users\jorge\Documents\Tec\Uni\6. Sexto Semestre\EEG_Luzma\EEG processing for NF syste
EEG = struct with fields:
             setname: ''
            filename: 'S01_eeglab.set'
            filepath: 'C:\Users\jorge\Documents\Tec\Uni\6. Sexto Semestre\EEG_Luzma\EEG processing for NF systems'
             subject: ''
               group: ''
           condition: ''
             session: []
            comments: ''
              nbchan: 14
              trials: 1
                pnts: 23040
               srate: 128
                xmin: 0
                xmax: 179.9922
               times: [0 7.8125 15.6250 23.4375 31.2500 39.0625 46.8750 54.6875 62.5000 70.3125 78.1250 85.9375 93.7
                data: [14×23040 single]
              icaact: []
             icawinv: []
           icasphere: []
          icaweights: []
         icachansind: []
            chanlocs: [1×14 struct]
          urchanlocs: []
            chaninfo: [1×1 struct]
                 ref: 'common'
               event: []
             urevent: []
    eventdescription: {}
               epoch: []
    epochdescription: {}
              reject: [1×1 struct]
               stats: [1×1 struct]
            specdata: []
          specicaact: []
          splinefile: ''
       icasplinefile: ''
              dipfit: []
             history: '↵EEG.etc.eeglabvers = '2025.1.0'; % this tracks which version of EEGLAB is being used, you m
               saved: 'justloaded'
                 etc: [1×1 struct]
                 run: []
             datfile: ''
                 roi: []
```

```
%Misma funcion que usa la GUI de eeglab, pero para script
```

Se pueden ver claramente los 14 canales

```
chan_num = EEG.nbchan
```

```
chan_num =
14
```

Canales nombres

```
chan_names = EEG.chanlocs
```

```
chan_names = 1×14 struct
```

...

Fields	labels	ref	theta	radius	X	Y	Z	sph_theta
1	'AF3'	''	[]	[]	[]	[]	[]	[]
2	'F7'	''	[]	[]	[]	[]	[]	[]
3	'F3'	''	[]	[]	[]	[]	[]	[]
4	'FC5'	''	[]	[]	[]	[]	[]	[]
5	'T7'	''	[]	[]	[]	[]	[]	[]
6	'P7'	''	[]	[]	[]	[]	[]	[]
7	'O1'	''	[]	[]	[]	[]	[]	[]
8	'O2'	''	[]	[]	[]	[]	[]	[]
9	'P8'	''	[]	[]	[]	[]	[]	[]
10	'T8'	''	[]	[]	[]	[]	[]	[]
11	'FC6'	''	[]	[]	[]	[]	[]	[]
12	'F4'	''	[]	[]	[]	[]	[]	[]
13	'F8'	''	[]	[]	[]	[]	[]	[]
14	'AF4'	''	[]	[]	[]	[]	[]	[]

El f3 es el 3 y el f4 es el 12

Sampling rate

```
fs = EEG.srate
```

```
fs =
128
```

La duracion es de 3 minutos. 6 historias con dos estados. Escuchando el relato del psicologo, recitandolo en su mente.

```
samples = EEG.pnts
```

```
samples =
23040
```

```
seconds = samples / fs
```

```
seconds =
180
```

```
minutes = seconds / 60
```

```
minutes =
3
```

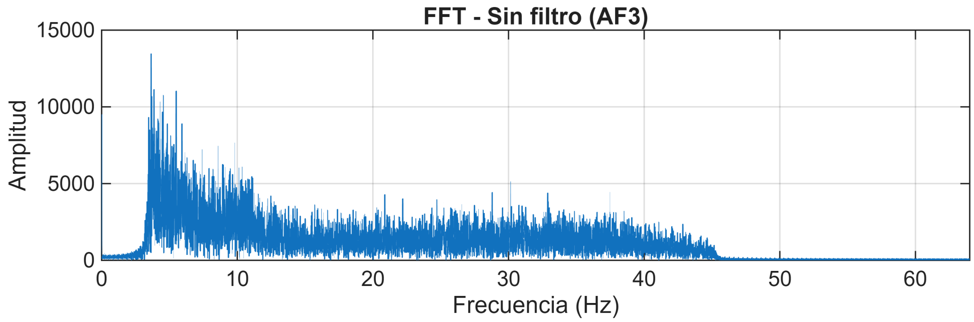
```
%Filtrado de señal
% Filtro pasa-bandas 1-45 Hz, elimina el drift y la señal de europa como de
% america. las ondas de interes (alpha) siguen estando presentes para el
% analisis.
EEG_filt = pop_eegfiltnew(EEG, 1, 45); %Es la misma funcion que usa la GUI de eeglab
```

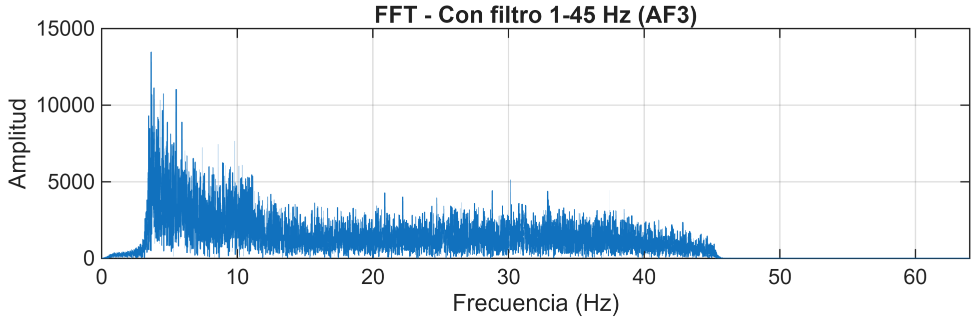
```
pop_eegfiltnew() - performing 425 point bandpass filtering.
pop_eegfiltnew() - transition band width: 0.99622641509434 Hz
pop_eegfiltnew() - passband edge(s): [1 45] Hz
pop_eegfiltnew() - cutoff frequency(ies) (-6 dB): [0.50188679245283 45.4981132075472] Hz
pop_eegfiltnew() - filtering the data (zero-phase, non-causal)
firfilt(): |====================| 100%, ETE 00:00
```

```
EEG_filt = eeg_checkset(EEG_filt);
```

Fourier de la señal filtrada y no filtrada

```
N  = EEG.pnts;
fs = EEG.srate;
f  = (0:N/2) * (fs/N);   % eje de frecuencia en Hz (solo positivo)

fourier_og  = abs(fft(EEG.data(1,:)));
fourier_new = abs(fft(EEG_filt.data(1,:)));

% Solo mitad positiva del espectro
mag_og  = fourier_og(1:N/2+1);
mag_new = fourier_new(1:N/2+1);

figure;
subplot(2,1,1);
plot(f, mag_og);
xlabel('Frecuencia (Hz)'); ylabel('Amplitud');
title('FFT - Sin filtro (AF3)');
xlim([0 64]); grid on;

subplot(2,1,2);
plot(f, mag_new);
xlabel('Frecuencia (Hz)'); ylabel('Amplitud');
title('FFT - Con filtro 1-45 Hz (AF3)');
xlim([0 64]); grid on;
```



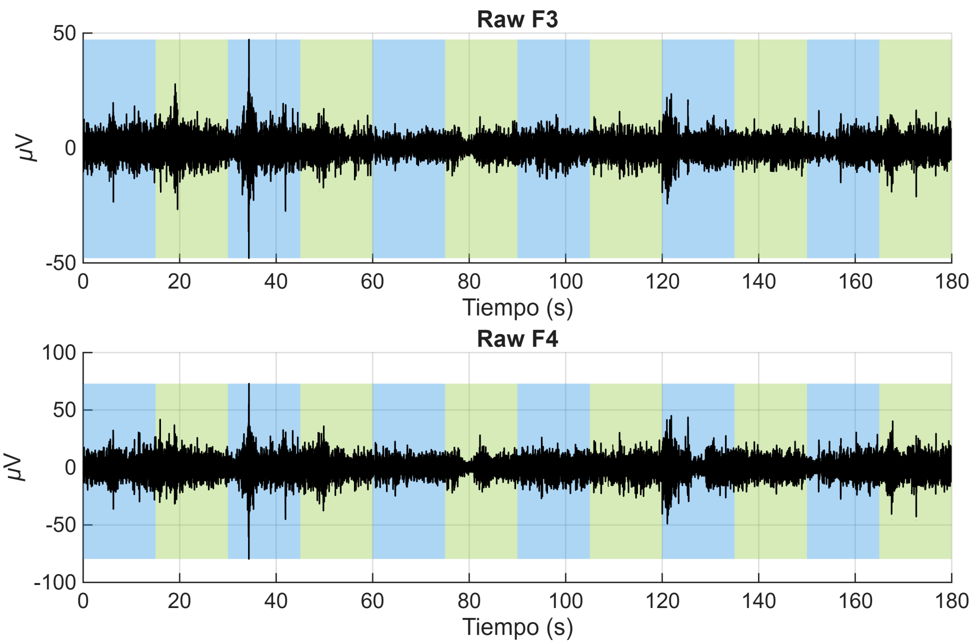


Podemos ver que si habia ruido de DC en 0 hz, pero no de linea (50-60 hz). Igualmente mejora la calidad. Probablemente por filtro interno del eeg que rechaza bandas que no nos son utiles en EEG.

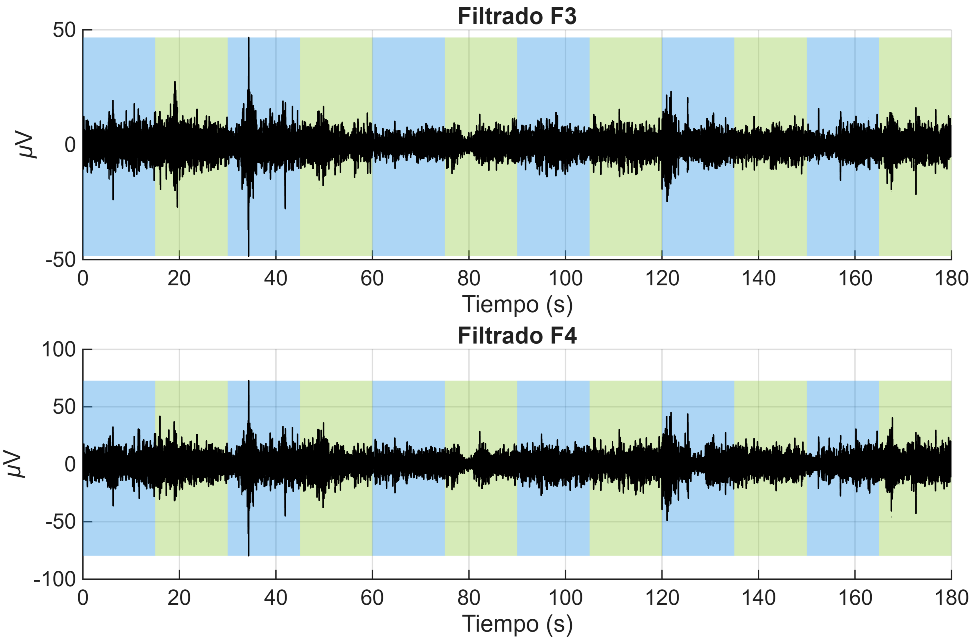
```
% Colores
color_listen = [0.2  0.6  0.9];
color_recall = [0.6  0.8  0.3];
t_axis = (0:EEG.pnts-1) / EEG.srate;
ch_names = {EEG.chanlocs.labels};
offset = 150;
```

```
t_axis = (0:EEG.pnts-1) / EEG.srate;
cl = [0.2 0.6 0.9];  cr = [0.6 0.8 0.3];   % Listen / Recall
raw  = EEG.data([3 12], :);
filt = EEG_filt.data([3 12], :);
ch_names = {'F3','F4'};

% --- Shading helper (reutilizado en las 3 figuras) ---
function shade(cl, cr, ylo, yhi)
    for i = 1:12
        fill([(i-1)*15 i*15 i*15 (i-1)*15],[ylo ylo yhi yhi], ...
             cl*(mod(i,2)==1)+cr*(mod(i,2)==0), ...
             'EdgeColor','none','FaceAlpha',0.4,'HandleVisibility','off');
    end
end

% --- Fig 1: Crudo ---
figure('Name','Raw F3 & F4');
```

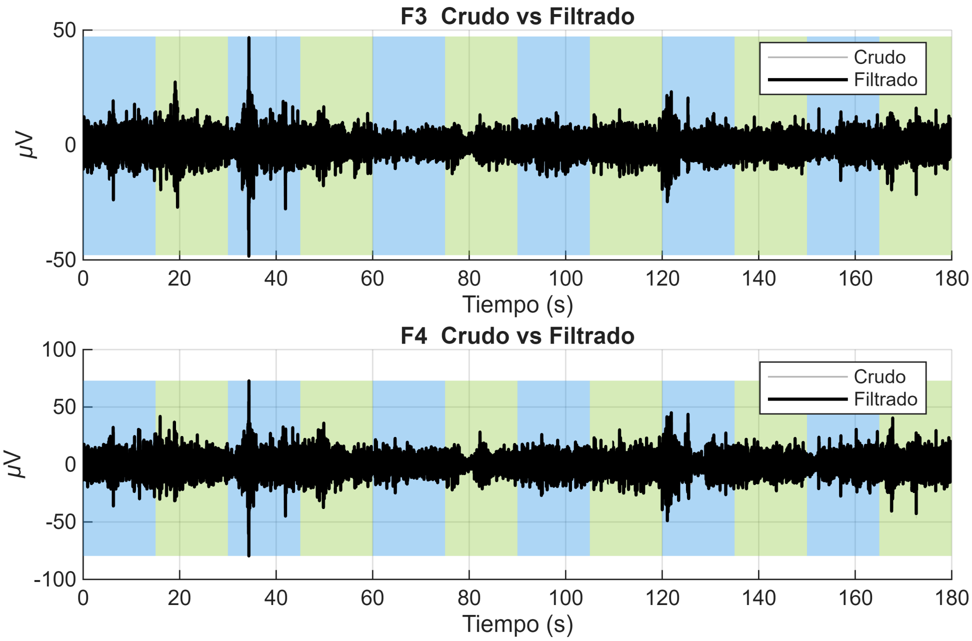
```
for k = 1:2
    subplot(2,1,k); hold on;
    shade(cl, cr, min(raw(k,:)), max(raw(k,:)));
    plot(t_axis, raw(k,:), 'k', 'LineWidth', 0.6);
    title(['Raw ' ch_names{k}]); xlabel('Tiempo (s)'); ylabel('\muV');
    xlim([0 180]); grid on; hold off;
end
```



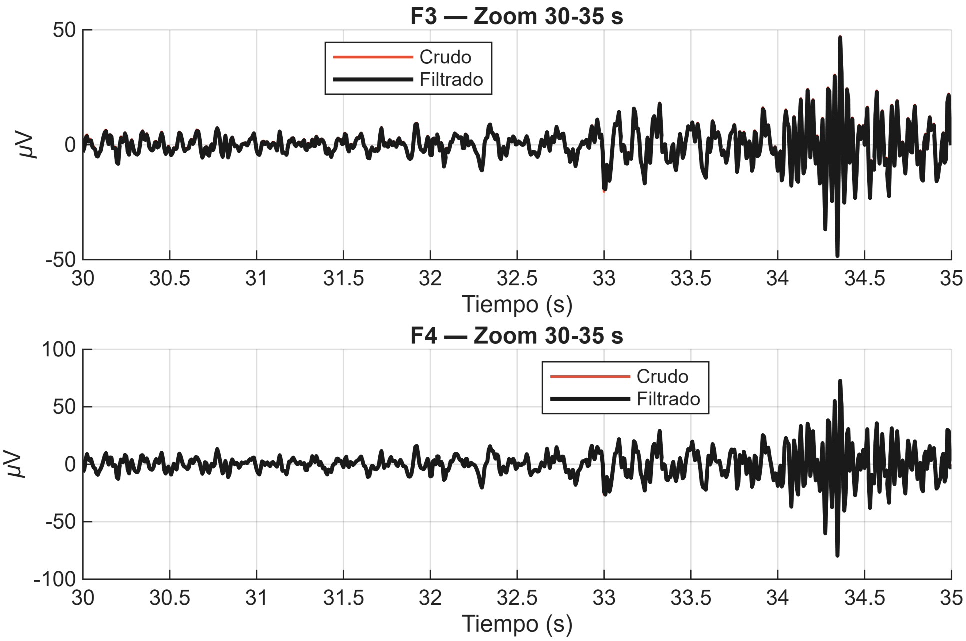

```
% --- Fig 2: Filtrado ---
figure('Name','Filtered F3 & F4');
for k = 1:2
    subplot(2,1,k); hold on;
    shade(cl, cr, min(filt(k,:)), max(filt(k,:)));
    plot(t_axis, filt(k,:), 'k', 'LineWidth', 0.6);
    title(['Filtrado ' ch_names{k}]); xlabel('Tiempo (s)'); ylabel('\muV');
    xlim([0 180]); grid on; hold off;
end
```



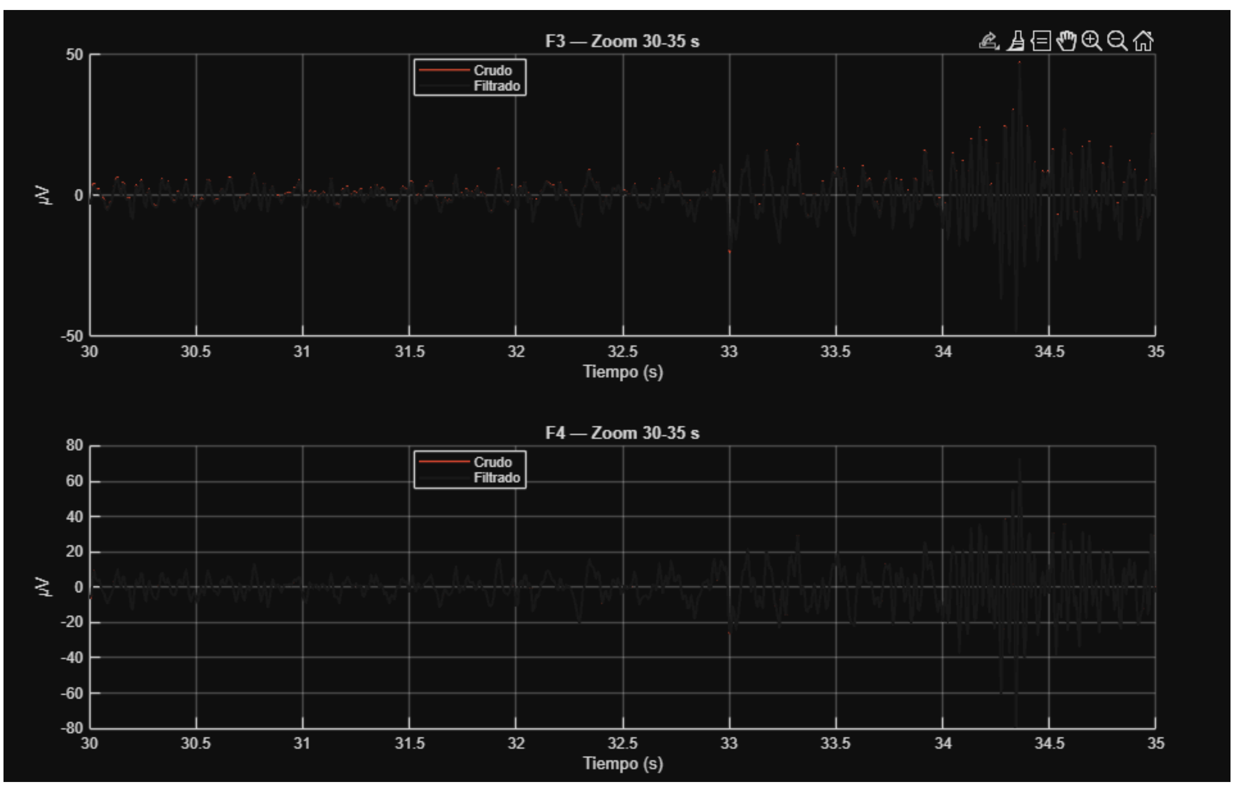

```
% --- Fig 3: Overlay crudo vs filtrado ---
figure('Name','Raw vs Filtered');
for k = 1:2
    subplot(2,1,k); hold on;
    shade(cl, cr, min(raw(k,:)), max(raw(k,:)));
    hr = plot(t_axis, raw(k,:),  'Color', [0.7 0.7 0.7], 'LineWidth', 0.6);
    hf = plot(t_axis, filt(k,:), 'k', 'LineWidth', 1.2);
    hr.Color(4) = 0.4;
    title([ch_names{k} '  Crudo vs Filtrado']);
    xlabel('Tiempo (s)'); ylabel('\muV');
    xlim([0 180]); grid on;
    legend([hr hf],'Crudo','Filtrado','Location','best');
    hold off;
end
```




```
figure('Name','Raw vs Filtered - Zoom');
zoom_range = t_axis >= 30 & t_axis <= 35;   % 5 segundos

for k = 1:2
    subplot(2,1,k); hold on;
    hr = plot(t_axis(zoom_range), raw(k,zoom_range),  'Color',[0.9 0.3 0.2],
'LineWidth', 1.0);
    hf = plot(t_axis(zoom_range), filt(k,zoom_range), 'Color',[0.1 0.1 0.1],
'LineWidth', 1.5);
    hr.Color(4) = 0.6;
    title([ch_names{k} ' — Zoom 30-35 s']);
    xlabel('Tiempo (s)'); ylabel('\muV'); grid on;
    legend([hr hf],'Crudo','Filtrado','Location','best');
    hold off;
end
```

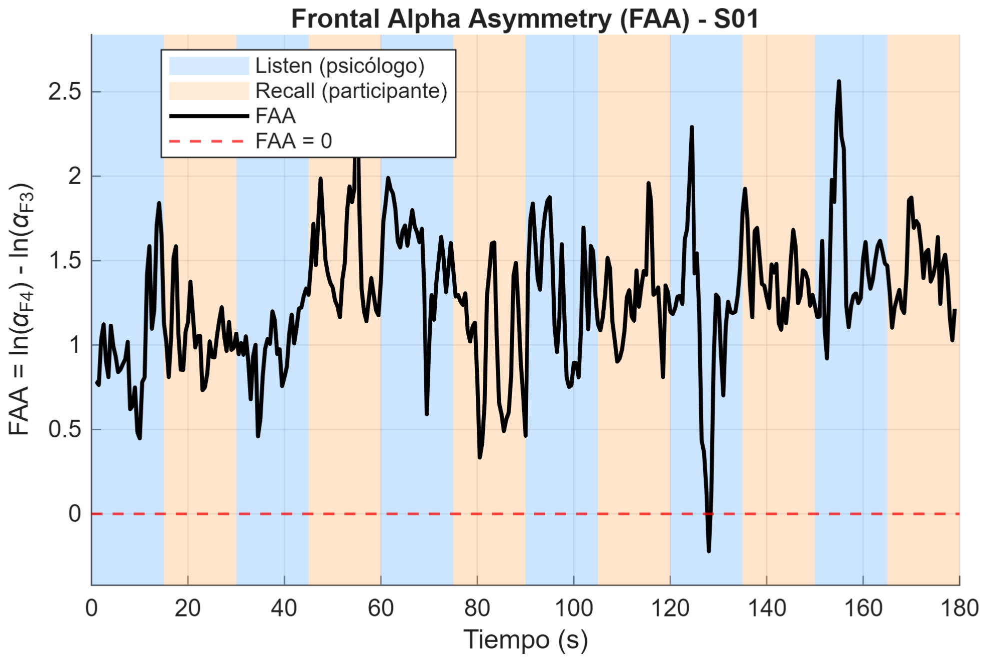


Como podemos ver en esta grafica el cambio despues del filtrado es muy sutil. pero si hay un cambio, porque como vimos en las graficas de fourier si habia potencia en frecuencia 0 hz. En el pdf exportado no se ve tambian asi que ademas adjunto una imagen



Frontal Alpha Assymetry

```matlab
%% FAA Neuromarker – F3/F4
fs = EEG_filt.srate;
win_samp = round(2 * fs);
step_samp = round(0.5 * fs);
alpha_low = 8; alpha_high = 13;

data_F3 = EEG_filt.data(3, :);
data_F4 = EEG_filt.data(12, :);

n_samp = size(EEG_filt.data, 2);
time_faa = []; faa_values = [];

for t = 1:step_samp:(n_samp - win_samp + 1)
    [pxx_F3, f] = pwelch(data_F3(t:t+win_samp-1), win_samp, win_samp/2, [], fs);
    [pxx_F4, ~] = pwelch(data_F4(t:t+win_samp-1), win_samp, win_samp/2, [], fs);
    idx = f >= alpha_low & f <= alpha_high;
    faa_values(end+1) = log(mean(pxx_F4(idx))) - log(mean(pxx_F3(idx)));
    time_faa(end+1) = (t + win_samp/2) / fs;
end

%% Plot
color_listen = [0.6 0.8 1.0];
color_recall = [1.0 0.8 0.6];

figure('Name','FAA Neuromarker');
ymin = min(faa_values) - 0.2; ymax = max(faa_values) + 0.2;
hold on;

for i = 1:12
    t0 = (i-1)*15; t1 = i*15;
    c = color_listen * (mod(i,2)==1) + color_recall * (mod(i,2)==0);
    fill([t0 t1 t1 t0], [ymin ymin ymax ymax], c, ...
        'EdgeColor','none', 'FaceAlpha',0.50, 'HandleVisibility','off');
end

h1 = fill(nan, nan, color_listen, 'FaceAlpha',0.4, 'EdgeColor','none');
h2 = fill(nan, nan, color_recall, 'FaceAlpha',0.4, 'EdgeColor','none');
h_faa = plot(time_faa, faa_values, 'k', 'LineWidth', 1.5);
h_zero = yline(0, '--r', 'LineWidth', 1);

xlabel('Tiempo (s)');
ylabel('FAA = ln(\alpha_{F4}) - ln(\alpha_{F3})');
title('Frontal Alpha Asymmetry (FAA) - S01');
legend([h1, h2, h_faa, h_zero], ...
    'Listen (psicólogo)', 'Recall (participante)', 'FAA', 'FAA = 0', ...
    'Location','best');
xlim([0 180]); ylim([ymin ymax]); grid on; hold off;
```




```
%% Estadísticas por fase
fprintf('FAA media Listen: %.4f\n', mean(faa_values(time_faa <= 90)));
```

```
FAA media Listen: 1.1913
```

```
fprintf('FAA media Recall: %.4f\n', mean(faa_values(time_faa > 90)));
```

```
FAA media Recall: 1.3417
```

La teoria de la FAA (Frontal alpha asymmetry) segun Davidson nos dice que mayor activacion frontal izquierda se asocia con estados de aproximacion y afecto positivo, en cambio mayor activacion frontal derecha nos indica evitacion y ansiedad.

Para calcularlo usamos una ventana deslizante de 2 seg con paso de 0.5 con overlap de 75%. Usamos pwelch en caad ventana. Extrajimos las frecuencias alfa de interes 8-13 hz y se le saca la media de la PSD.

Finalmente se resta el logaritmo natural de lo que ya habiamos sacado de la media. Derecho- izquierdo.